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Effects of Waterlogging at Different Stages of Development on the Growth and Yield of Peas (*Pisum sativum* L.)

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The effects of waterlogging, for up to 8 days at different stages of development, on the growth and yield of peas have been examined. The peas were grown outdoors subject to field conditions in lysimeters containing monolith sandy loam or clay soils, in facilities where drainage and rainfall could be controlled. The experiments, made during 2 years, involved waterlogging treatments likely to be the most extreme occurring in the pea-growing areas of the UK. During each waterlogging treatment the oxygen concentration at 20 cm depth in the soil declined to less than 2% within 2 days. On average, when the soil was waterlogged to the surface for 5 days at the 1–2, 3–4 and 6–7 leaf stages (the latter being just before flowering), the losses of yield of seeds (peas) at the freezing stage were 6, 15 and 42%, respectively. The severity of the effect depended on the duration of the treatment and the height of the water-table; at the 6–7 leaf stage either 2 days waterlogging to the surface or 5 days with the water-table at 50 cm below the soil surface had little effect on seed yield. Waterlogging restricted leaf expansion and internode extension. After waterlogging at the pre-flowering stage, leaves usually senesced prematurely and the shoot apex became prematurely quiescent. In all experiments the yield of haulm was more affected by waterlogging than were the yields of pods and seeds. The losses in yield were mainly a consequence of fewer pods per plant. Waterlogging before flowering for 5 days depressed the concentrations of nitrogen, phosphorus and potassium in the haulm grown on both soils and of nitrogen in the seeds (peas) of plants in 1 year on the clay. Other treatments had little effect on nutrient concentration. After waterlogging to the surface for 5 days before flowering the uptake of water by roots was restricted, but other treatments had only a small effect, being least in the early treatments.

1. Introduction

The case has been made for obtaining information on the effects of temporary waterlogging on crop growth to improve the basis of the design of field drainage systems,[1] and as a first step the effects of waterlogging on winter wheat at different stages of growth have been reported previously.[2] Results for peas (*Pisum sativum* L.) are presented here.

Peas are known to be susceptible to anaerobic soil conditions[3,4] and, in a preliminary investigation,[5] waterlogging for 24 h or more just before flowering restricted growth and yield; the effects of waterlogging at earlier and later stages were less marked. In those experiments the peas were grown with wide spacing and little interplant competition, and plants were supported by stakes whereas, in the experiments described here, the peas were grown with cultural conditions similar to those usually found in the field.

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2. Experimental

The plants were grown in 1977 and 1979 in undisturbed monoliths of soil contained in lysimeters, with facilities to control rainfall and drainage but with conditions otherwise similar to those in the field. There were 32 lysimeters (80 cm in diameter and 135 cm deep) for each of two soils, a sandy loam and a clay. A guard crop was grown in soil between and around the lysimeters. Full details of the facilities and soils have been published previously.[1,6]

The waterlogging treatments (Table 1) were selected as representing the most severe conditions likely to prevail in the areas of the UK where peas are grown, and were imposed by a combination of natural rainfall and irrigation (Table 2). The main features of the agronomy (Table 3) were similar to those for field crops. In each experiment the same widely grown early variety of vining pea (cv. Sprite) was grown. In 1979 on the sandy loam, a nematicide (oxamyl at 2.5 kg a.i. ha^{-1}) was applied before sowing to control the free-living eelworms (*Trichodorus* spp.) in this soil.[2] In each year winter wheat preceded the peas.

Table 1. Experimental treatments and design

Experiment	Treatments and growth stage of waterlogging[a]	Starting date	Duration (days)	Experimental design
1977, Clay	Freely-drained			
	3–4 leaves	6 May	5	Two Latin squares
	3–4 leaves	6 May	8	Eight replicates
	6–7 leaves	30 May	5	
1979, Clay	Freely-drained			
	1–2 leaves	16 April	5	Two Latin squares
	3–4 leaves	5 May	5	Eight replicates
	6–7 leaves	19 May	5	
1979, Sandy loam	Freely-drained			
	3–4 leaves	3 May	5	Randomised block
	6–7 leaves	19 May	2	Six replicates
	6–7 leaves	19 May	5	(Eight on freely-drained)
	6–7 leaves	19 May	5[b]	

[a] Waterlogging to soil surface unless indicated.
[b] Water-table at 50 cm below soil surface.

Table 2. Monthly and total water applied (mm) during experiments (the same amount of water was applied to all lysimeters in each experiment, with natural rainfall supplemented by irrigation where necessary)

	February	March	April	May	June	July	*Total*
1977	55	25	48	126	47	36	337
1979							
Clay	21	29	124	193	22	19	408
Sandy loam	21	29	91	281	22	19	463

Table 3. Main agronomic details of experiments

Experiment	Sowing date	Plants[a] (m^{-2})	Fertiliser (kg K ha^{-1})	Harvest date
1977	21 March	114	nil	11 July
1979	6 March	96	clay—nil	2 July
1979	6 March	96	sandy loam—60	2 July

[a] 130 seeds m^{-2} were planted; there were no significant differences in plant populations between treatments.

During the experiments the oxygen concentration of the soil atmosphere or soil water during waterlogging was estimated as described previously.[2] The time course of the water content of the soil profile was measured using a neutron moisture meter, with the access tube located in the centre of each lysimeter. Soil temperature at 20 cm was measured with thermistors connected to a data logger.

In both years the growth stage of the plants was assessed frequently as the basis for timing of the waterlogging treatments. In 1979 vegetative growth was measured on five marked plants on each lysimeter. Leaf area during growth was calculated by multiplying the product of the measured length and breadth by a factor of 0.72, determined from measurements with a Paton Electronic Planimeter on leaves excised from plants in the guard area.

The numbering of nodes, internodes and leaves was in accordance with the system described by Lovell,[7] where leaf 1 is found at node 3, with nodes 1 and 2 bearing trifid bracts. Most plants formed their first flowers at node 10, i.e. in the axil of leaf 8, but a small proportion started flowering at node 9.

The marked plants were harvested separately; internode lengths and the area of fully green leaves were measured and the numbers of pods and seeds per pod at each flowering node were counted. The fresh and dry weights of pods, seeds and haulm were recorded and added to the corresponding weights of the other plants on the lysimeters.

It was intended to harvest the seeds (peas) when they were ready for freezing, as in commercial conditions; this corresponds to a tenderometer reading of 100. In the lysimeters there were insufficient plants for repeated tenderometer readings. In the 1977 experiment the timing of harvest was therefore assessed on the visual appearance of the peas. In the 1979 experiments timing was aided by measuring the percentage of alcohol-insoluble solids[8] in the seeds, since this is an index of tenderness, with 12% corresponding to a tenderometer reading of 100 (Bingham, J., Processors and Growers Research Organisation, Peterborough, private communication). The measurements were first made on samples of seeds taken from the guard area, at increasing frequency as harvest approached, and then on small samples from lysimeters of each treatment (their weight being added to that of the remaining plants for yield determination). A further check on the appropriateness of the harvest date was made by calculating the accumulated heat units (for peas a heat unit is every °C between a 'base temperature' of 4.4°C for crop growth and the mean of the daily minimum and maximum air temperatures[9]). The number of heat units for cv. Sprite to reach a tenderometer reading of 100 is on average 689;[9] this was reached in the 1979 experiments on 2 July. In 1977 the plants were harvested after 731 heat units had accumulated, i.e. nearer the canning stage (*ca* 733 heat units[9]), when the tenderometer reading was 120; the freezing stage would have been about 4 days earlier.

At harvest in both years the concentrations of nitrogen, phosphorus and potassium in the haulm, pods (fruit minus seeds) and seeds were measured, using methods described previously.[2]

Table 4. Mean soil temperature (°C) at 20 cm during each waterlogging treatment and at 30 cm, 1941–70 for Oxfordshire and West Berkshire[10]

	Stage of waterlogging		
	1–2 leaves (April)	3–4 leaves (early May)	6–7 leaves (late May)
Temperatures during experiments (mean of 12 readings each day)			
1977		10.9 for 5 days 11.3 for 8 days	13.6
1979	10.4	8.8	12.4

Mean soil temperature at 30 cm			
	April	May	June
	8.4	11.8	15.4

3. Results

3.1. Soil temperature

The soil temperature at 20 cm was compared with the long-term average at 30 cm, the best available comparison.[10] In 1977 it was close to the long-term mean (Table 4). In 1979 the soil was relatively warm at the 1–2 leaf stage, relatively cool at the 3–4 leaf stage and subsequently close to the average.

3.2. Soil oxygen concentrations

As in previous experiments with these soils in spring, the oxygen concentration declined rapidly after the start of waterlogging to less than 2 volumes % at 20 cm depth within 2–3 days. As an example, some results for 1977 in the clay soil are shown (Figure 1). At 50 cm the oxygen concentration declined more slowly (to 6%) and the difference between the waterlogged and freely-drained soil was less than at 20 cm.

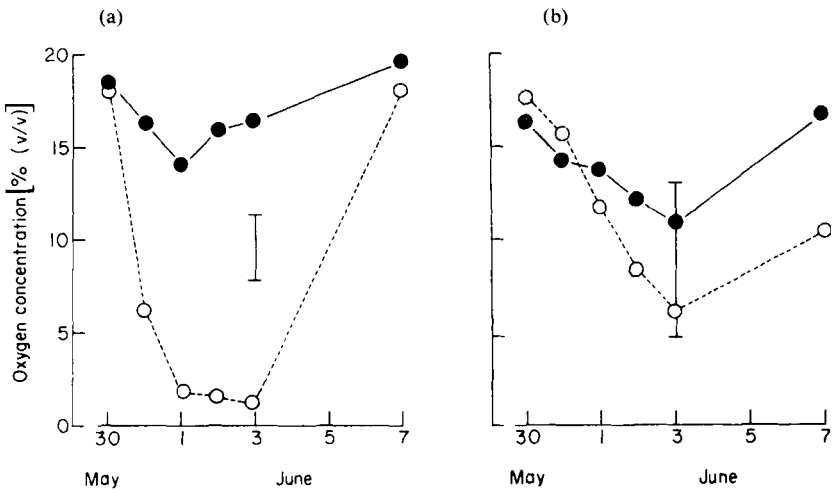


Figure 1. Effect of waterlogging on the concentration of oxygen in a clay soil at (a) 20 cm; and (b) 50 cm depth. Waterlogging started on 30 May and drainage began on 4 June 1977. ———, freely-drained treatment; – – – –, waterlogged treatment. Vertical bars denote least significant difference (LSD) at $P=0.05$.

3.3. Crop growth and yield

3.3.1. 1977 clay soil

The plants were disease-free throughout the experiment. During waterlogging the plants were not obviously damaged but, 3 weeks after waterlogging at the 3–4 leaf stage, leaves on the treated plants were markedly chlorotic; leaf area and plant height were depressed by 42 and 14%, respectively, after 5 days' waterlogging, and by 69 and 24% after 8 days of the treatments. After the pre-flowering waterlogging the older leaves became prematurely senescent, but the younger leaves were less affected than in the earlier treatments.

At harvest the total yield of the plants was significantly depressed by all the waterlogging treatments, especially by the 8-day treatment starting at the 3–4 leaf stage; this depressed the yield of seeds by 30% (Table 5). After 5 days of waterlogging at that time, or before flowering at the 6–7 leaf stage, the yield of seeds was depressed by only 13 and 10%, respectively. For the 8-day vegetative waterlogging and the pre-flowering treatment the yield of haulm was proportionately more affected than that of seeds (Table 5).

The loss in yield by the 8-day treatment was associated with both fewer pods per plant and fewer seeds per pod (Table 5). The concentrations of nitrogen, phosphorus and potassium in the haulm and seeds were similar in all treatments.

3.3.2. 1979 clay soil

Vegetative growth

Internode extension was restricted by waterlogging (Table 6). The effect was least in plants waterlogged at the 1–2 leaf stage, being significantly only for internode 10. On plants waterlogged at the 3–4 leaf stage the last two internodes before flowering and the first four internodes after flowering started, were significantly shortened. On plants waterlogged at the 6–7 leaf stage, just before flowering, all the internodes after flowering were shortened.

Leaf expansion was also restricted by waterlogging, the effect being more immediate and pronounced than that on internode extension (Table 7). For plants waterlogged at the 1–2 leaf stage, leaves 3–7 were smaller, but leaf 8 was similar in area to that on the freely-drained treatment.

Premature leaf senescence was observed as a result of waterlogging at the 3–4 and 6–7 leaf stages, but not on plants waterlogged at the 1–2 leaf stage. The first signs of yellowing were noticed 4–7 days after the water-tables were lowered. The leaves affected were those fully expanded when the waterlogging started (Table 8). On plants waterlogged at the 6–7 leaf stage the shoot apex senesced prematurely, contributing to the shorter length of the plants (Table 6). The pods already formed when the apices became quiescent matured at the same rate as on the other treatments.

Table 5. Effects of waterlogging on yield and components of yield; clay soil, 1977 (figures in parentheses are waterlogged treatments as % of freely-drained)

		Waterlogged			
	Freely-drained	3–4 leaves 5 days	3–4 leaves 8 days	6–7 leaves 5 days	LSD ($P<0.05$)
Dry matter yield (t ha^{-1})					
Total shoot	10.82	9.49	6.89	8.68	1.00
Haulm	4.93	4.26 (86)	2.81 (57)	3.66 (74)	0.47
Pods	2.43	2.22 (91)	1.67 (69)	1.91 (79)	0.24
Seeds (peas)	3.46	3.01 (87)	2.41 (70)	3.11 (90)	0.48
Pods per plant	5.3	5.4	4.6	4.5	0.6
Seeds per pod	5.1	4.6	4.5	5.2	0.4
Weight per seed (mg)	110	104	102	115	8

Table 6. Effects of waterlogging on the final internode length (mm) of pea plants grown on clay soil, 1979

			Waterlogged for 5 days			
	Internode	Freely-drained	1–2 leaves	3–4 leaves	6–7 leaves	LSD ($P<0.05$)
	3	9	9	—	—	0.6
	4	10	*9	—	—	1.4
	5	21	17	—	—	2.5
	6	28	28	*30	—	2.4
	7	31	29	27	—	3.4
	8	42	38	32	—	3.4
	9	66	58	48	*62	8.1
Flowering begins						
	10	87	77	63	62	7.3
	11	80	76	64	48	9.8
	12	85	84	74	35	11.2
	13	85	81	72	22	10.2
	14	57	55	50	13	14.8
	15	24	25	22	6	13.2
	16	7	5	5	<1	7.9

* Stage when each waterlogging treatment started.

Table 7. Effects of waterlogging for 5 days on area of green leaves (cm^2) of pea plants grown on clay, 1979

Leaf number	Freely-drained	Waterlogged			LSD ($P < 0.05$)
		1–2 leaves	3–4 leaves	6–7 leaves	
1	4.9	4.7	—	—	0.6
2	6.8	*5.7	—	—	0.7
3	9.4	6.2	8.9	—	1.0
4	14.9	9.6	*10.0	—	1.8
5	18.5	10.8	9.8	—	1.4
6	20.8	15.8	9.6	—	2.2
7	30.9	25.8	12.4	*22.7	4.1
8	38.1	37.6	18.9	18.3	6.8

* Stage when each waterlogging treatment started.

Table 8. Effects of waterlogging for 5 days on the % of senesced leaves on pea plants on clay soil; 30 May 1979

Leaf number	Freely-drained	Waterlogged		
		1–2 leaves	3–4 leaves	6–7 leaves
1	25	10	70	100
2	15	10	70	100
3	0	5	15	100
4	0	0	0	95
5	0	0	0	80
6	0	0	0	55
7	0	0	0	0

Yield

All waterlogging treatments depressed yields of haulm, pods and seed. The effects were more pronounced the later the treatment occurred (Table 9). For each treatment the haulm weight was more affected than that of the pods or seed. Waterlogging at the 1–2 leaf stage did not have a significant effect on any yield components and waterlogging at the 3–4 leaf stage did not significantly affect seed yield, but after waterlogging at the 6–7 leaf stage yields of haulm, pods and seeds were approximately halved.

The lighter seed yield was a consequence largely of fewer pods per plant. The number of pods on the first three flowering nodes was unaffected by the treatments, but plants waterlogged at the 6–7 leaf stage had fewer pods on the later flowering nodes (Table 10), so that there were almost two fewer pods per plant on this treatment. The mean weight of individual seeds was slightly less after waterlogging at the 6–7 leaf stage than on the freely-drained treatment but the effect was not significant (Table 9). The heaviest seeds were in the oldest pods and although the mean weight of seeds at each node tended to be less on the plants waterlogged at the 6–7 leaf stage the effect was not significant (Table 10).

The concentrations of nitrogen, phosphorus and potassium in the haulm and of nitrogen in the seeds were depressed by the pre-flowering waterlogging, but not by treatment at earlier stages (Table 11).

3.3.3. 1979 sandy loam soil

Vegetative growth

The effects on internode extension and on leaf expansion of waterlogging to the soil surface at a particular growth stage was much the same as on the clay soil. For plants waterlogged to the soil surface at the 6–7 leaf stage for only 2 days the length of the last three internodes also was significantly shorter (Table 12), but where the water-table had been at 50 cm at this growth stage the

Table 9. Effects of waterlogging peas for 5 days on yield and components of yield; clay soil, 1979 (figures in parentheses are waterlogged treatments as % of freely-drained)

	Freely-drained	Waterlogged			LSD ($P<0.05$)
		1–2 leaves	3–4 leaves	6–7 leaves	
Dry matter yield (t ha^{-1})					
Total shoot	7.93	7.09	5.54	4.03	1.55
Haulm	3.90	3.39 (87)	2.48 (64)	1.89 (48)	0.70
Pods	2.00	1.80 (90)	1.44 (72)	1.08 (54)	0.37
Seeds (peas)	2.03	1.90 (94)	1.62 (80)	1.05 (52)	0.62
Pods per plant	5.2	4.8	4.5	3.2	0.7
Seeds per pod	4.6	4.8	4.4	4.5	0.4
Weight per seed (mg)	83	85	86	78	15

Table 10. Effects of waterlogging on the number of pods and the mean seed weight at each node; clay soil, 1979

Node number	Freely-drained	Waterlogged			LSD ($P<0.05$)
		1–2 leaves	3–4 leaves	6–7 leaves	
Number of pods					
10	0.57	0.59	0.54	0.63	0.36
11	1.03	1.29	0.93	0.97	0.31
12	1.25	1.10	1.09	0.98	0.37
13	1.05	0.88	1.16	0.49	0.33
14	0.94	0.61	0.99	0.11	0.32
15	0.30	0.38	0.52	0.13	0.30
Mean seed weight (mg)					
10	141	129	121	110	38
11	111	121	114	100	18
12	101	105	104	86	18
13	91	93	86	71	20
14	70	68	72	55	29
15	62	49	57	40	22
16	43	50	50	—	18
17	27	—	—	—	—

Table 11. Effects of waterlogging on the concentration (mg g^{-1}) of nitrogen, phosphorus and potassium in haulm and seeds (peas); clay soil, 1979

	Freely-drained	Waterlogged			LSD ($P<0.05$)
		1–2 leaves	3–4 leaves	6–7 leaves	
Haulm					
N	23.1	25.0	24.8	20.6	3.5
P	2.9	3.0	3.0	1.9	0.5
K	19.4	19.8	19.8	15.3	2.4
Seeds (peas)					
N	41.6	41.2	40.3	38.6	1.4
P	4.8	4.8	4.8	5.0	0.3
K	10.6	10.9	10.8	10.8	0.5

elongation of internodes was not affected. Leaves senesced prematurely only on plants waterlogged to the surface for 5 days at the 6–7 leaf stage. The mean leaf areas per plant at harvest were 154 and 121 cm^2 for the freely-drained and 3–4 leaf stage waterlogging treatments and 93, 42 and 141 cm^2 for the treatments waterlogged at the 6–7 leaf stage for 2 days, 5 days and with the water-table at 50 cm depth.

Yield

As on the clay soil, waterlogging for 5 days at the 3–4 and 6–7 leaf stages significantly depressed the yield of haulm; the largest effect followed the later treatment (Table 13). Waterlogging for 2 days at the 6–7 leaf stage caused a loss in yield similar to that following 5 days waterlogging at the 3–4 leaf stage. With both these treatments the yield of haulm was more affected than that of pods or seeds. When the water-table was raised only to 50 cm from the soil surface at the 6–7 leaf stage yields of haulm, pods and seeds were not significantly affected (Table 13).

The main factor accounting for the lighter yield of seed was the smaller number of pods per plant (Table 12), but after waterlogging to the soil surface at the 6–7 leaf stage for 5 days mean seed weight was also much less than in the other treatments. The number of pods at each node in that treatment was significantly less after the second flowering node, and the mean weight of the seeds significantly less at all nodes (Table 14).

The concentrations of nitrogen and phosphorus in the haulm, but not in the seeds were depressed by waterlogging to the surface for 5 days before flowering, but not by the other treatments (Table 15).

3.4. Soil water content

The total soil moisture deficits in the top 100 cm of the profile at harvest in 1979 were expressed relative to field capacity after the last waterlogging treatment (Table 16). The deficits were largest on the freely-drained treatments on both soils; the later the stage of growth when waterlogging occurred, the smaller the soil moisture deficit at harvest. At all depths less water was extracted by the roots of plants that were waterlogged at the later stages of growth although the differences were significant only for the 6–7 leaf waterlogging treatment. Results for the sandy loam are given in Figure 2; similar results were obtained in the clay soils in 1977 and 1979.[11]

Table 12. Effects of waterlogging on the final internode length (mm) of pea plants grown on a sandy loam soil, 1979

		Waterlogged				
Internode	Freely-drained	3–4 leaves	6–7 leaves 2 days	6–7 leaves 5 days	6–7 leaves 5 days with water-table at 50 cm	LSD ($P<0.05$)
3	9					
4	11	*				
5	21	25				2
6	28	31				3
7	32	31				3
8	40	37				3
9	59	50	*	*	*	9
Flowering begins						
10	79	68	85	64	80	8
11	79	69	77	43	84	6
12	80	72	72	31	79	6
13	80	64	64	14	73	10
14	53	41	39	2	45	11
15	22	17	6	—	20	10
16	—	2	—	—	5	7

* Stage when each waterlogging treatment started.

Table 13. Effects of waterlogging on yield and components of yield; sandy loam, 1979 (figures in parentheses are waterlogged treatments as % of freely-drained)

		Waterlogged				
	Freely-drained	3–4 leaves	6–7 leaves 2 days	6–7 leaves 5 days	6–7 leaves 5 days with water-table at 50 cm	LSD ($P<0.05$)
Dry matter yield (t ha^{-1})						
Total shoot	8.18	6.65	6.82	3.49	7.47	1.50
Haulm	3.84	2.90 (76)	2.97 (77)	1.84 (48)	3.32 (86)	0.59
Pods	2.30	1.92 (83)	1.89 (82)	0.99 (43)	1.98 (86)	0.47
Seeds (peas)	2.05	1.83 (89)	1.96 (96)	0.66 (32)	2.17 (106)	0.58
Pods per plant	5.4	4.6	4.7	2.5	5.4	0.7
Seeds per pod	4.6	4.7	5.0	4.3	5.0	0.6
Weight per seed (mg)	82	85	88	63	91	17

Table 14. Effects of waterlogging peas on the number of pods and the mean seed weight at each node; sandy loam, 1979

		Waterlogged				
Node number	Freely-drained	3–4 leaves	6–7 leaves 2 days	6–7 leaves 5 days	6–7 leaves 5 days with water-table at 50 cm	LSD ($P<0.05$)
Number of pods						
10	0.45	0.64	0.40	0.67	0.43	0.36
11	0.86	0.92	1.20	1.29	0.76	0.40
12	1.38	1.14	1.25	0.68	1.16	0.38
13	1.12	1.19	1.20	0.31	1.15	0.34
14	0.98	0.62	0.60	0.03	1.00	0.32
15	0.35	0.46	0.10	—	0.72	0.31
Mean seed weight (mg)						
10	136	116	134	82	135	39
11	119	109	116	77	139	28
12	117	95	110	57	125	33
13	96	88	95	45	125	30
14	68	86	79	—	100	28
15	59	73	86	—	80	34
16	39	65	—	—	51	58
17	25	—	—	—	27	2

4. Discussion

Peas are generally susceptible to wet soil conditions,[12] but it is thought that the present results are the first to show the effect on yield from waterlogging in field conditions at different stages of growth.

The crop yields obtained were typical of those in commerce[9] but the losses in yield from waterlogging were less pronounced than in experiments in glasshouses[4] or outdoors in reconstituted soil profiles and with unusually wide plant spacing.[5] In these last experiments, individual plants were staked and were much taller than usual, with more nodes and with three to four more pods per plant; also the individual seeds at harvest were heavier, reflecting the more advanced state of maturity at that time, and there was a correspondingly greater contrast between the weights of seeds on freely-drained and waterlogged plants.

The reasons for the greater resistance of younger pea plants to waterlogging (Table 9) are uncer-

Table 15. Effects of waterlogging on the concentration (mg g^{-1}) of nitrogen, phosphorus and potassium in haulm and seeds (peas); sandy loam, 1979

		Waterlogged				
	Freely-drained	3–4 leaves	6–7 leaves 2 days	6–7 leaves 5 days	6–7 leaves water-table at 50 cm 5 days	LSD ($P<0.05$)
Haulm						
N	23.4	22.5	22.4	19.3	21.0	3.9
P	2.4	2.4	2.2	1.7	2.3	0.5
K	14.7	14.3	11.7	13.9	12.7	3.2
Seeds (peas)						
N	41.8	42.1	41.2	40.4	41.0	1.8
P	4.8	4.8	4.6	5.1	4.8	0.3
K	11.2	11.2	10.9	12.0	11.1	0.4

Table 16. Soil moisture deficits to a depth of 100 cm in clay and sandy loam soils, when peas were harvested, 1979

	Soil moisture deficit (mm)	
Treatment	Evesham clay	Skipwith sandy loam
Freely-drained	93.4	108.9
Waterlogged		
1–2 leaf stage, 5 days	82.5	—
3–4 leaf stage, 5 days	73.5	92.7
6–7 leaf stage, 2 days	—	81.6
6–7 leaf stage, 5 days	45.0	32.8
6–7 leaf stage, 5 days to 50 cm	—	90.3
LSD ($P<0.05$)	16.4	20.7

tain, but such plants do have some ability to regenerate new roots after waterlogging whereas older plants do not.[5] The effect of waterlogging on the ability to extract water from the soil was greater in older than in younger plants (Table 16; Figure 2), perhaps because of damage to the roots or because stomata had closed. Other possible explanations for the greater resistance of young plants to waterlogging have been suggested by Jackson.[4] However, in the experiment on the clay soil in 1977 waterlogging at the 6–7 leaf stage caused no greater effect than at the 3–4 leaf stage (Table 5).

Vegetative growth was affected less in these experiments than in the experiments in more artificial environments. For example, premature senescence of leaves was observed only in the most damaging waterlogging treatments (Table 8) rather than in all of them,[4, 5] and no desiccation of leaves even in the more extreme treatments was observed.

Restricted extension of internodes on waterlogged plants was common to these experiments and those in more artificial environments. On the clay soil, after waterlogging at the 1–2 leaf stage internode extension was significantly restricted only at the beginning of flowering (Table 6). Waterlogging at the 3–4 leaf stage on both soils did not affect shoot extension until internode 9 (just before flowering), but it also affected the four internodes after flowering (Tables 6 and 12). Waterlogging at the pre-flowering 6–7 leaf stage shortened the length of all subsequent internodes. Leaf expansion was much more rapidly affected by waterlogging, being suppressed by the end of the treatment (Table 7). The effect on leaf expansion of waterlogging at the 1–2 leaf stage seemed to have disappeared by the leaf stage 7–8.

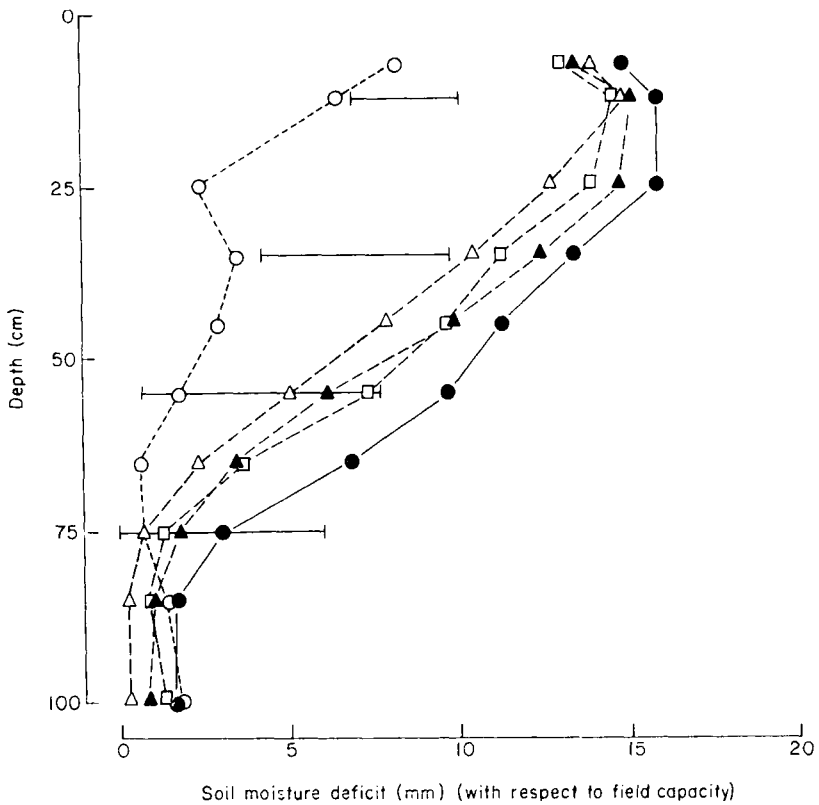


Figure 2. Soil moisture deficits in a sandy loam on 2 July (at harvest), developed after the 6–7 leaf stage waterlogging treatments started on 19 May 1979; at that time lysimeters that were freely-drained or had been waterlogged at the 3–4 leaf stage were returned to field capacity. ●, freely-drained; ▲, waterlogged for 5 days 3–4 leaf stage, water-table at soil surface; △, waterlogged for 2 days 6–7 leaf stage, water-table at soil surface; ○, waterlogged for 5 days 6–7 leaf stage, water-table at soil surface; and □, waterlogged for 5 days 6–7 leaf stage, water-table at 50 cm. Horizontal bars denote least significant difference (LSD) at $P=0{\cdot}05$.

The effect of waterlogging in restricting the number of fruiting nodes (Tables 10 and 14) was more evident the later the treatment and was most severe with the pre-flowering waterlogging. Thus, the main factor accounting for the lighter yield after the later waterlogging treatments (Tables 10 and 14) was that there were fewer pods on the younger nodes.

In all experiments, waterlogging affected the yield of haulm more than that of pods or seeds (Tables 5, 9 and 13). This may be partly because photosynthate from the pods themselves is of major importance in the growth of seeds.[13,14] Only the most damaging waterlogging treatments depressed nutrient uptake; the effects in the haulm were greater than in the seeds (Tables 11 and 15), as also noted by Jackson.[4]

Several species of grain legumes are sensitive to waterlogging[15,18] such that only a short period of it can cause large losses of yield. The species concerned, *Cajanus indicus* Spreng, *Vigna unguiculata* (L.) Walp, *Phaseolus vulgaris* L. and *Glycine max* (L.) Merrill, are all adapted to warmer environments than peas, yet when peas were waterlogged for similar short periods in these experiments similar losses of yield were found as in the other species. Since the sensitivity of plants (including legumes[19]) to waterlogging often increases with temperature, relatively small effects might have been expected on peas grown in temperature conditions. Not all temperate species of grain legumes are so badly affected by waterlogging; in the UK field beans (*Vicia faba*) are traditionally sown in autumn on clay soils where anaerobic conditions may prevail for long periods in winter.

Most experiments investigating the effects of transient waterlogging have used the most extreme

treatment, i.e. waterlogging to the soil surface. In practice less extreme waterlogging would be more common. In soya beans, raising the water-table for 7 days to 45 cm below the surface had little effect when the plants were growing vegetatively, but depressed yields by 40% when it was done after the pods had set;[20] after waterlogging at the earlier stages (but not later) some regrowth of the root system occurred. In the experiments on the sandy loam, raising the water-table to within 50 cm of the soil surface for 5 days just before flowering slightly depressed the yield of haulm and pods but not of seeds (Table 13), probably because most of the root system was in the upper part of the soil profile. This may, therefore, represent about the maximum degree of waterlogging that the crop should suffer at that stage. The crop can withstand higher water-tables for a shorter period; waterlogging to the surface at that stage for 2 days significantly depressed the yield of haulm, but not the yield of peas (Table 13).

The seed yield losses following waterlogging to the soil surface for 5 days at the 1–2, 3–4 and 6–7 leaf stages were on average 6, 15 and 42%, respectively. These losses are much greater than would be shown by winter wheat if waterlogged for similar periods.[2] Waterlogging of wheat for 6 days in late May caused a loss of 0.2 t ha^{-1} (valued £20 at 1979 prices). By contrast, 5 days of waterlogging of peas at that time depressed yield by about 0.9 t ha^{-1} (Tables 5, 9 and 12) valued at about £250. Although 25 days' waterlogging of wheat in April depressed yield by about 1.25 t ha^{-1}, waterlogging peas for such a period would almost certainly kill the crop. Results such as these can assist in deciding the intensity of drainage to install for different cropping systems, and they strongly indicate the desirability of avoiding growing peas on soil types where, without a major investment in drainage, waterlogging would be common.

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References

1. Cannell, R. Q.; Belford, R. K.; Gales, K.; Dennis, C. W. A lysimeter system used to study the effect of transient waterlogging on crop growth and yield. *J. Sci. Food Agric.* 1980, **31**, 105–116.
2. Cannell, R. Q.; Belford, R. K.; Gales, K.; Dennis, C. W.; Prew, R. D. Effects of waterlogging at different stages of development on the growth and yield of winter wheat. *J. Sci. Food Agric.* 1980, **31**, 117–132.
3. Erickson, A. E.; Van Doren, D. M. The relations of plant growth and yield to oxygen availability. *Proc. 7th Int. Congr. Soil Sci.* Madison, Wisconsin, 1960, IV, **54**, 429–434.
4. Jackson, M. B. Rapid injury to peas by soil waterlogging. *J. Sci. Food Agric.* 1979, **30**, 143–152.
5. Cannell, R. Q.; Gales, K.; Snaydon, R. W.; Suhail, B. A. Effects of short-term waterlogging on the growth and yield of peas (*Pisum sativum*). *Ann. appl. Biol.* 1979, **93**, 327–335.
6. Belford, R. K. Collection and evaluation of large soil monoliths for soil and crop studies. *J. Soil Sci.* 1979, **30**, 363–373.
7. Lovell, P. H. Correlative influences in seedling growth. In *The Physiology of the Garden Pea* (Sutcliffe, J. F.; Pate, J. S., Eds), Academic Press, London, 1977, pp. 265–290.
8. Houlbrooke, A. The differentiation of canned processed and canned garden peas. *J. Assn. of Public Analysts* 1963, **1**, 16–18.
9. Gane, A. J.; King, J. M.; Gent, G. P. *Processing and Growers Research Organisation Handbook* Vol. 1—*Peas*. PGRO, Peterborough, 1971.
10. Smith, L. P. The agricultural climate of England and Wales. Ministry of Agriculture, Fisheries and Food. *Technical Bulletin 35* HMSO, London, 1976, p. 147.
11. Cannell, R. Q. Effects of soil drainage on root growth and crop production. In *Soil Physical Properties and Crop Production in the Tropics* (Lal, R.; Greenland, D. J., Eds), John Wiley and Sons, Chichester, 1979, pp. 184–197.
12. Van Hoorn, J. W. Results of a ground water level experimental field with arable crops on clay soil. *Neth. J. Agric. Sci.* 1958, **6**, 1–10.
13. Milbourn, G. M.; Hardwick, R. C. The growth of vining peas. I. The effect of time of sowing. *J. Agric. Sci. Camb.* 1968, **70**, 393–402.
14. Flinn, A. N.; Pate, J. S. A quantitative study of carbon transfer from pod and subtending leaf to the ripening seeds of the field pea (*Pisum arvense* L.). *J. exp. Bot.* 1970, **21**, 71–82.

15. International Crops Research Institute for the Semi-Arid Tropics. Pigeon pea breeding. *ICRISAT Annual Report 1975–76* 1976, 93.
16. Minchin, F. R.; Summerfield, R. J.; Eaglesham, A. J. R.; Stewart, K. A. Effects of short-term waterlogging on growth and yield of cowpea (*Vigna unguiculata*). *J. Agric. Sci. Camb.* 1978, **90**, 355–366.
17. Forsythe, W. M.; Victor, A.; Gomez, M. Flooding tolerance and surface drainage requirements of *Phaseolus vulgaris* L. In *Soil Physical Properties and Crop Production in the Tropics* (Lal, R.; Greenland, D. J., Eds), John Wiley and Sons, Chichester, 1979, pp. 205–214.
18. Wien, C.; Lal, R.; Pulver, E. L. Effects of transient flooding on growth and yield of some tropical crops. In *Soil Physical Properties and Crop Production in the Tropics* (Lal, R.; Greenland, D. J., Eds), Chichester, 1979, pp. 235–245.
19. Heinricks, D. H. Flooding tolerance of legumes. *Can. J. Pl. Sci.* 1970, **50**, 435–438.
20. Stanley, C. D. Soybean top and root response to static and fluctuating water-table situations. Ph.D. dissertation, Iowa State University, Ames, Iowa, 1978.